Cardiologist & Healthcare Administrator - Analytics Dashboard Journey

Project Overview

- **Project Title:** Heart Disease Analysis & Patient Risk Trends[cite: 25]
- **Repository:** Heart-Disease-Analysis-Dashboard[cite: 25]
- **Website:** Flask web application with an embedded Tableau Dashboard and Tableau Story[cite: 25]
- **Team Leader:** Mounika Sanapathi[cite: 25]
- **Team Members:** Thakasi Balaji, Kale Sai Praneeth, Navyasri Gonnuri, Shanthi Kumari Challa[cite: 25]

Project Description

The project analyzes approximately 50,000 patient clinical and health records using Tableau Public[cite: 25]. It delivers vital clinical insights through an interactive Tableau Dashboard and Tableau Story, both embedded in a Flask-based web application[cite: 25].

Core Dashboard Analysis Areas

- **Clinical KPI Dashboard:** Total patient count, high-risk case distributions, and average patient survival rates[cite: 25].
- **Demographic Distribution:** Breakdown of heart disease prevalence by age groups and gender[cite: 25].
- **Biometric Risk Factor Breakdown:** Multi-variable correlation of cholesterol levels, resting blood pressure, and fasting blood sugar[cite: 25].
- **Lifestyle Impact Analysis:** Correlation of smoking habits, physical activity levels, and dietary habits with cardiovascular risk[cite: 25].
- **Diagnostic Test Analysis:** Insights from electrocardiogram (ECG) results and maximum heart rate achieved during stress tests[cite: 25].
- **Chest Pain Type Classification:** Distribution and severity tracking of angina types[cite: 25].
- **Geographical Patient Patterns:** Regional data analyzing external environmental or regional influences on heart health[cite: 25].
- **Interactive Diagnostic Filters:** Ability to isolate patient data by specific health risk intersections[cite: 25].

Objective: To assist cardiologists, healthcare administrators, and clinical data analysts in recognizing early-stage cardiovascular risk patterns, optimizing hospital resource allocation, and executing data-driven medical strategies[cite: 25].

Customer Persona

Primary User

- **Role:** Chief Cardiologist / Healthcare Medical Director[cite: 25]
- **Age Range:** 35–55 years[cite: 25]
- **Core Responsibility:** Oversees daily clinical operations, patient treatment guidelines, and high-level medical outcomes within the cardiology department[cite: 25].

Core Goals

- Understand complex multi-variable patient risk profiles[cite: 25].
- Monitor critical clinical key performance indicators and patient intake volume[cite: 25].
- Identify high-prevalence cardiovascular trends to implement targeted preventative care[cite: 25].
- Improve overall patient survival rates and long-term health outcomes[cite: 25].
- Make data-backed decisions regarding diagnostic protocols and clinical staffing[cite: 25].

Operational Challenges

- Managing massive volumes of raw, siloed Electronic Health Records (EHR)[cite: 25].
- Difficulty manually tracking subtle, interconnected physiological correlations[cite: 25].
- Limited real-time visibility into overall hospital patient risk trends[cite: 25].
- Time-consuming, labor-intensive clinical and administrative reporting processes[cite: 25].

Patient Analytics Journey Map

Stage	1. Access Dashboard	2. Review Clinical KPIs	3. Analyze Risk Behavior	4. Review Insight Story	5. Execute Decisions
Experience Steps	Opens the Flask clinical portal and loads the embedded Tableau interface[cite: 25].	Scans medical KPI cards for patient metrics, high-risk percentages, and blood markers[cite: 25].	Filters data by age, chest pain type, lifestyle indicators, and clinical history[cite: 25].	Walks through the guided Tableau story for epidemiological insights[cite: 25].	Applies insights to preventative health programs and hospital resource decisions[cite: 25].
Interactions	Flask application homepage with native Tableau dashboard integration[cite: 25].	Summary metrics and dynamic risk-stratification overview charts[cite: 25].	Interactive drop-downs, cross-filtering, and diagnostic drill-down charts[cite: 25].	Multi-layered, slide-by-slide historical patient data narrative[cite: 25].	Clinical board meetings, staff briefings, and exported medical PDF reports[cite: 25].
Goals & Motivations	Get an immediate, high-level snapshot of current patient demographic distributions[cite: 25].	Understand whether patient risk metrics are stable, rising, or falling[cite: 25].	Isolate specific co-morbidities and identify highly vulnerable patient profiles[cite: 25].	See the overarching epidemiological picture rather than isolated metrics[cite: 25].	Implement data-backed clinical guidelines and preventative strategies with confidence[cite: 25].
Positive Moments	Dashboard loads rapidly with an intuitive, clean medical UI layout[cite: 25].	Risk-stratified KPI cards make complex patient data instantly comprehensible[cite: 25].	Deep filters expose unexpected correlations between lifestyle and test results[cite: 25].	The visual narrative effortlessly translates raw statistics into actionable health trends[cite: 25].	Clinical leadership feels secure knowing strategies are backed by thousands of case files[cite: 25].

Negative Moments	The density of medical metrics makes it initially unclear where to look first[cite: 25].	Certain health KPIs lack baseline clinical context (normal vs. abnormal control metrics)[cite: 25].	Overwhelming combinations of filters can complicate fast decision-making during shifts[cite: 25].	The insights can feel static if the underlying EHR database is not refreshed frequently[cite: 25].	Hesitance to finalize hospital-wide systemic protocol updates without peer review[cite: 25].
Areas of Opportunity	Introduce a brief, interactive "Clinician Onboarding Guide" for first-time dashboard users[cite: 25].	Integrate standard clinical control baselines and color-coded alert thresholds (Red/Yellow/Green) [cite: 25].	Create pre-set, smart default filters based on high-priority medical demographics[cite: 25].	Establish automated weekly or monthly data syncs with the hospital's central patient registry[cite: 25].	Generate automated, shareable summaries containing direct data-driven medical recommendations[cite: 25].

Prepared as part of the Heart Disease Analysis and Patient Risk Trends capstone project[cite: 25].